

TED (15) -6092

Reg.no.....

(REVISION—2015)

Signature.....

MODEL QUESTION PAPER

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/MANAGEMENT/MANAGEMENT/COMMERCIAL PRACTICE

QUALITY ASSURANCE AND TESTING

(Maximum marks: 100)

Time: 3 hours

PART—A

(Maximum marks: 10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define flexible plastics.
2. Ammonium sulphate is used for the coagulation of latex in VFA number determination .Give reason.
3. Define M300 and how it is expressed.
4. Name two basic tests for water tanks.
5. Define Brittleness temperature .

(5*2=10)

PART—B

(Maximum marks: 30)

II. Answer any five of the following questions. Each question carries 6 marks.

1. Explain the procedure for determination of TSC of NR latex.
2. Write down the technical specification for ISNR as per BIS.
3. Define Tension set and explain the method to determine it.
4. Explain two specification tests for cycle rubber tube .
5. Describe the procedure for determination of dielectric constant of polymers .
6. Describe the procedure to determine the density of NR latex .
7. Explain with a diagram of MFI tester, the determination of MFI for plastics. How is it related to molecular weight?

(5*6=30)

PART—C

(Maximum marks: 60)

(Answer one full Question from each unit. Each full question carries 15 marks)

UNIT--I

- III (a) Describe the functions of standards organizations- DIN and ASTM. 8
(b) Describe the procedure for determination of percentage purity of DPG and MBT.7

OR

- IV (a) Describe the procedure to determine arc resistance of polymers. 7
(b) Determine the basic setup and procedure to determine Vicat softening temperature 8
Of polymers .

UNIT—II

- V (a) Describe the procedure to determine dirt content of ISNR 8
(b) Describe the procedure to determine sludge content of concentrated NR latex. 7

OR

- VI (a) Define VFA number. Explain the significance and procedure to determine VFA Number.8
(b) Describe the significance and procedure to determine magnesium content of NR latex. 7

UNIT—III

- VII (a) Define abrasion resistance and explain the procedure to determine it using Dupont abrader . 7
(b) Explain the procedure to determine scorch time using mooneyviscometer . 8

OR

- VIII (a) Define Tensile strength and explain how it can be determined using a tensile testing machine. 8
(b) Explain the determination of cure characteristics using ODR.7

UNIT—IV

- IX (a) Describe the method to determine leakage test and tensile properties of Cycle rubber Tube. 8

- (b) Describe two test methods for LPG tubings.7

OR

- X (a) Describe the method to determine Split tear strength and Hardness of Rubber Hawaiichappal8
(b) Describe the Indentation hardness test for latex foam rubber. 7
